



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Department of Computer Science
Faculty of Science and Technology

CSC 2210: OBJECT ORIENTED PROGRAMMING 2

Spring 2025-2026

Section: W

Group No:

Project Report On MCQ Vault

Supervised By
Al Jubair Hossain

Submitted By:

Name	ID
1. Shajadul Islam Shopno	25-61255-1
2. Jannatul Naima Champa	25-61263-1
3. Apurba Rahman	24-57037-1
4. Marufa Ahsan Preety	25-61283-1

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				Total =
Requirement fulfillment	Properly demonstrate a real-life scenario-based project with proper functional requirement identification for the Object-Oriented Programming project development activities.				
Validation	Ensuring the ability of students' proper demonstration on validation forms in their system in terms of dealing with the data.				
Verification	Identifying if the students can verify the system data along with proper functional requirements in terms of data flow.				

Project Content

- 1)Introduction
- 2)Pictorial View
- 3)Designing and Diagrams
- 4)Author Part
- 5)Examiner Part
- 6)Benefits of MCQ Vault Project in practical Scenarios
- 7)In Future Project Upgrade Planning
- 8)Conclusion

Introduction

MCQ Vault is a project developed by .NET 7 windows form application(Winforms) where students answer to Multiple Choice question created by Authors in order to make themselves better at analytical thinking, sharpening basics and solving conceptual questions .

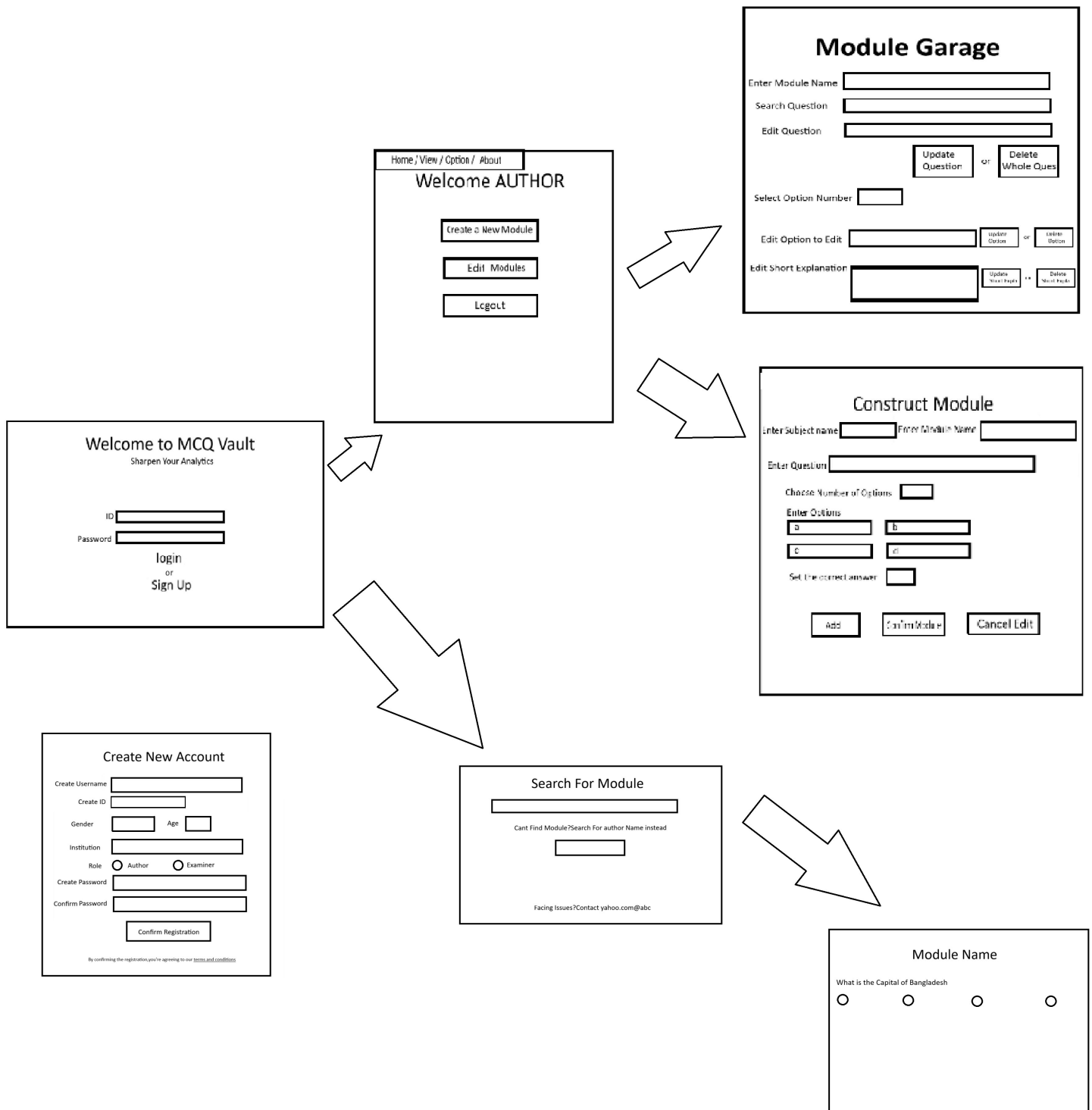
In this project Multiple students can interact with multiple types of modules. A module consists of number of questions that are hosted by Authors . Each Author have different identity . Students have to search for module name in order to attend the practice examination .

The system was designed with multiple diagrams . Initially in this report, a pictorial demonstration is added in order to get basic impression about the project . After that Multiple diagrams were drawn for example ER Diagram for Database, Use case diagram for visualization of actors and automated systems ,Class Diagram for the System sturcture and lastly state chart Diagram for the Dynamic Behaviour of the System .

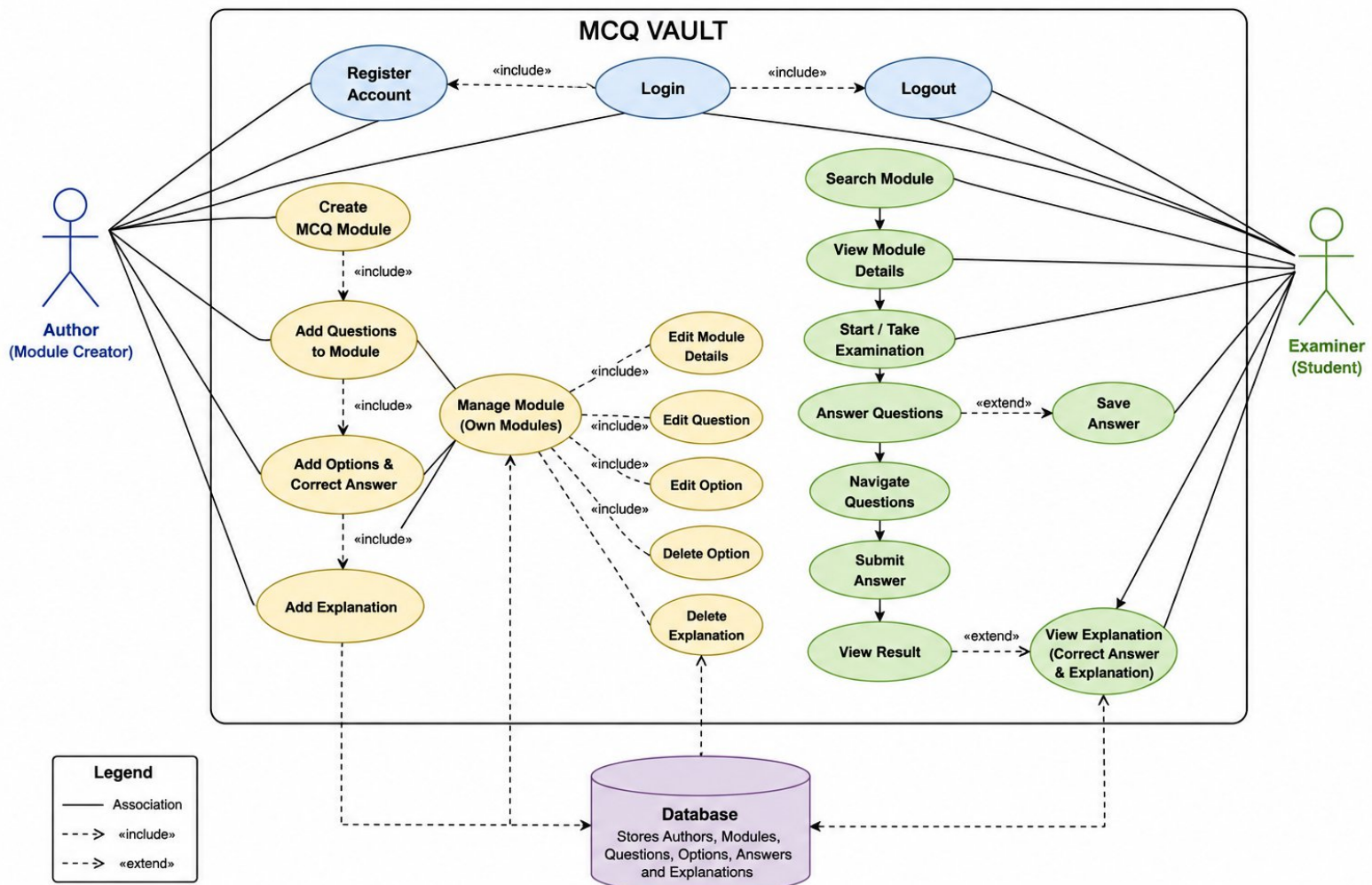
After Diagram Demonstration,pictures of every form were added of both Examiner and Author Part .

At the end some discussions for example benefits of this project,future planning and Conclusion were added in order to complete an ending of a report .

Pictorial View



Use Case Diagram



ER Diagram

AUTHOR		
Int	Author_Id	PK
string	Author_Name	
string	Gender	
string	Institution	
string	Author_Password	

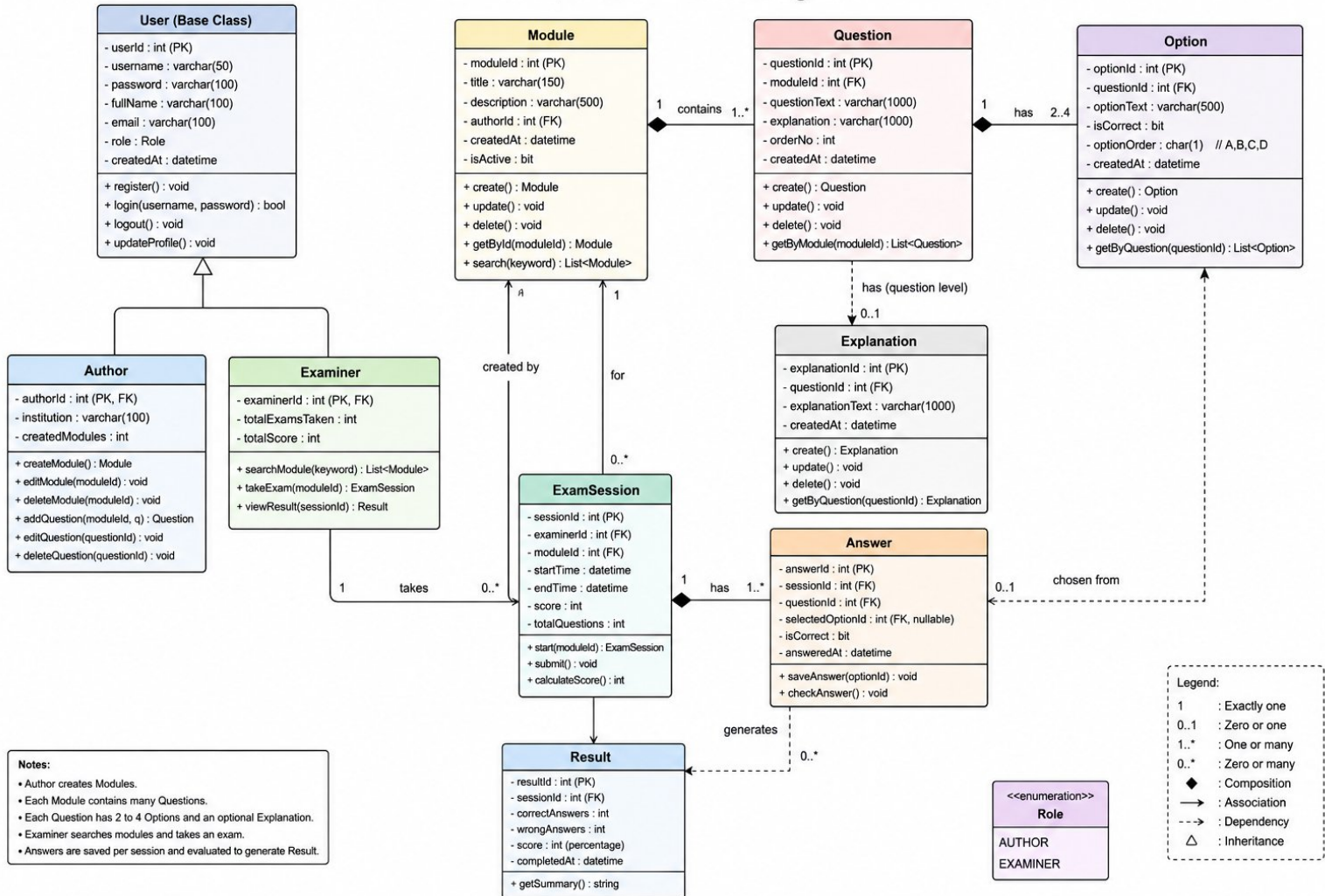


EXAMINER		
Int	Examiner_Id	PK
string	Examiner_Name	
string	Gender	
string	Institution	
string	Examiner_Password	

MODULE		
string	Module_Name	
string	Question	
string	Option_A	
string	Option_B	
string	Option_C	
string	Option_D	
string	Correct_Answer	
string	Short_Explanation	
Int	Module_HostId	FK

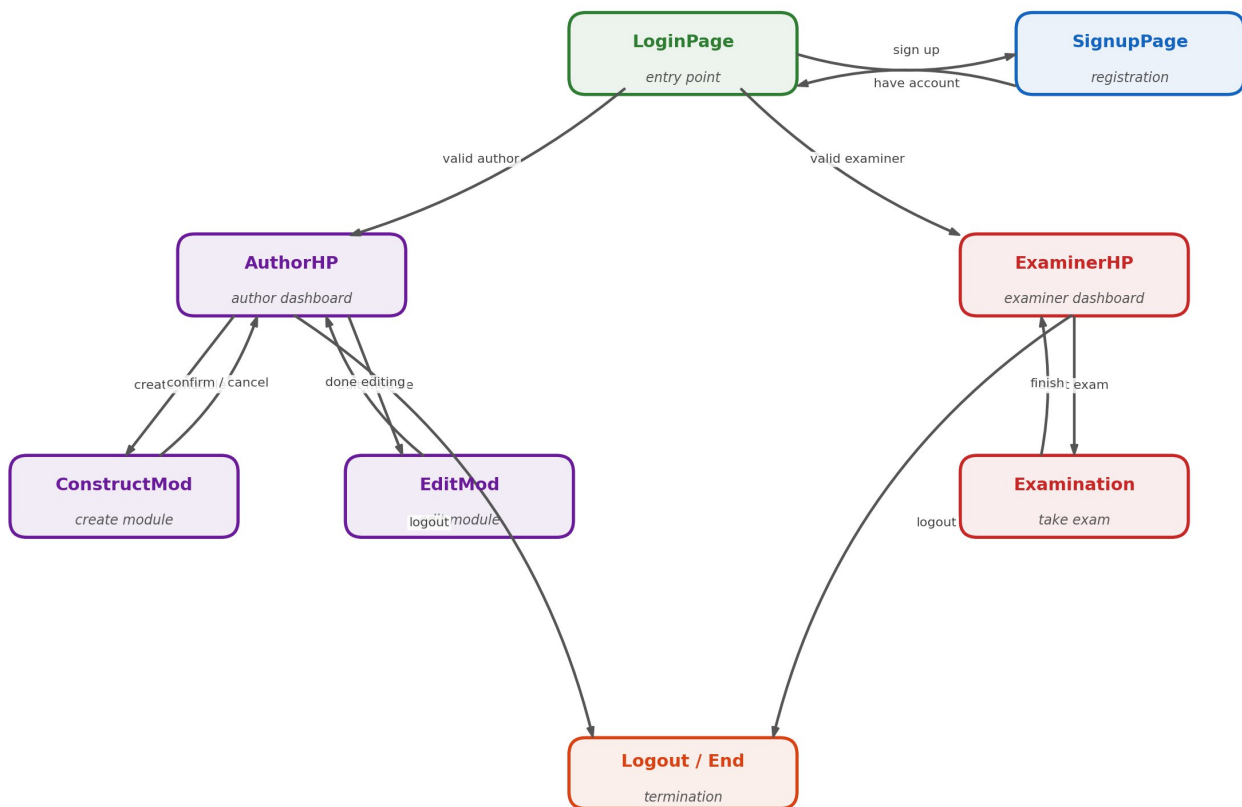
Class Diagram

MCQ VAULT – Class Diagram



State Chart Diagram

MCQ Vault — State Chart Diagram



Entry state

Author flow

Examiner flow

Auth / shared

Termination

Login Session

MCQ Vault

Welcome to MCQ Vault

Sharpen Your Basics and Analytics

Username

Password

Or

[Don't Have an account ? Click Here to Sign up instead](#)

SignupPage

Create New Account

Create Username

Gender

Institution

Role ☐ Author ☐ Examiner

Create Password

Password Must contain more than 8
Password must contain UPPERCASE , lowercase and numbers

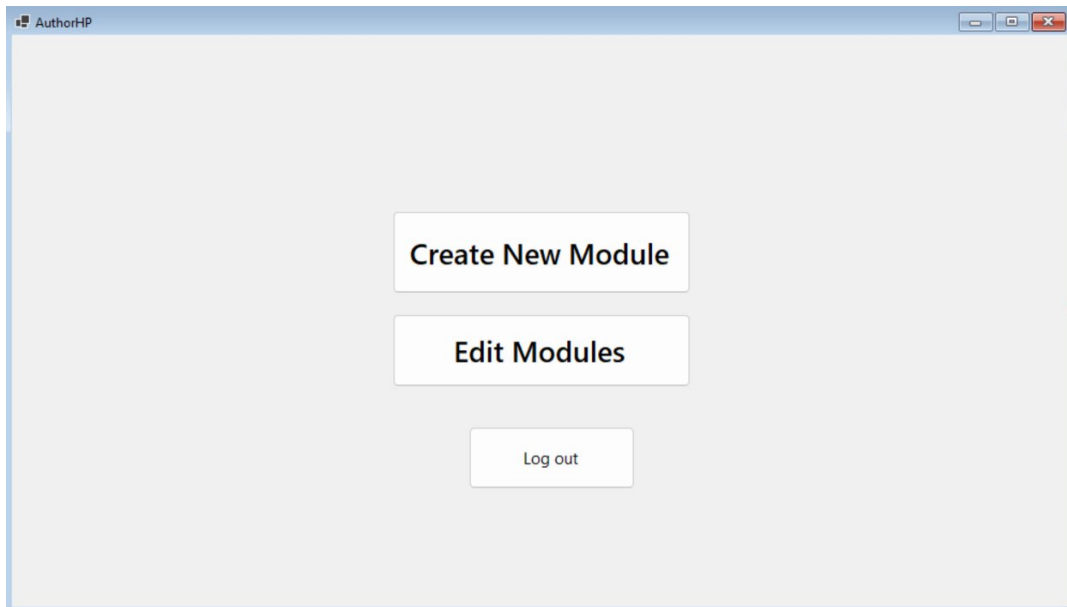
Confirm Password

Password Didn't Matched .Recheck matchings

[Have An Account? Login Instead](#)

By Confirming the registration,you're agreeing to our [terms and conditions](#)

Author Design



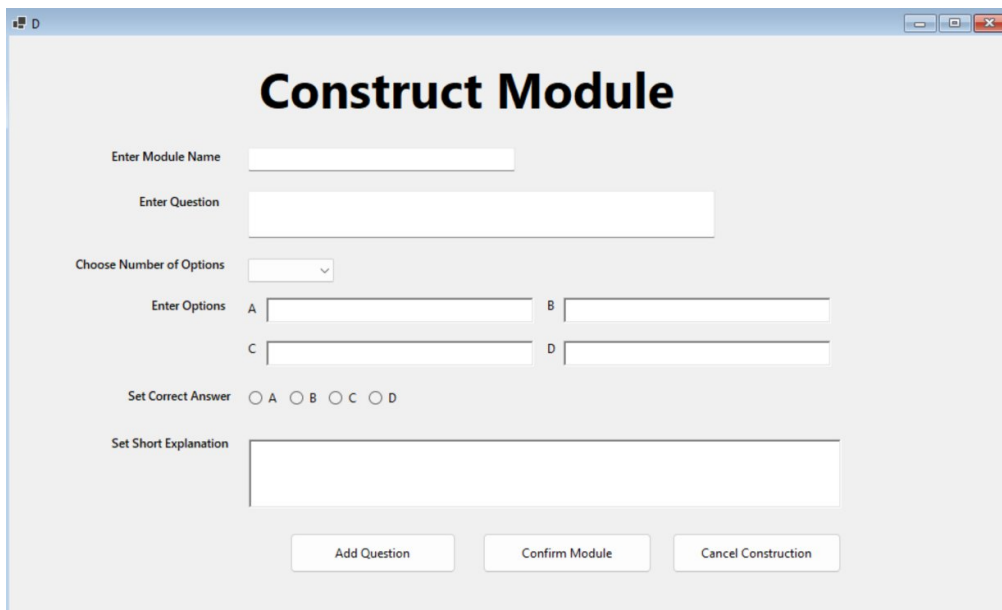
AuthorHP

Create New Module

Edit Modules

Log out

This is the main menu of the AuthorHP application. It features three large, white buttons with black text, arranged vertically in the center of a light gray background. The buttons are labeled 'Create New Module', 'Edit Modules', and 'Log out'. The window has a standard Windows-style title bar with the text 'AuthorHP' and window control icons.



Construct Module

Enter Module Name

Enter Question

Choose Number of Options

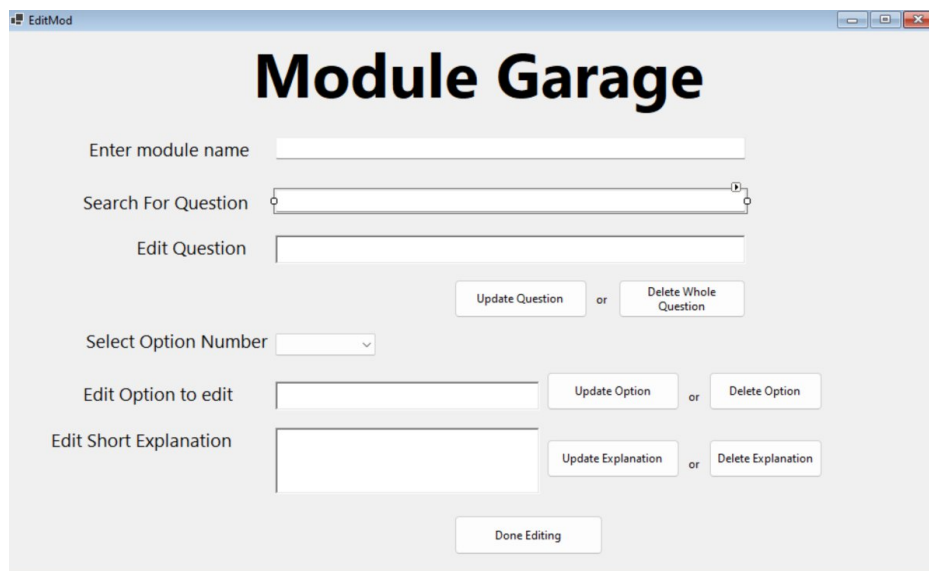
Enter Options A B
C D

Set Correct Answer ☐ A ☐ B ☐ C ☐ D

Set Short Explanation

Add Question Confirm Module Cancel Construction

This is the 'Construct Module' form. It has a title bar with 'D'. The form is titled 'Construct Module' in bold. It contains several input fields: 'Enter Module Name', 'Enter Question', 'Choose Number of Options' (a dropdown menu), 'Enter Options' (four text boxes labeled A, B, C, and D), 'Set Correct Answer' (radio buttons for A, B, C, and D), and 'Set Short Explanation' (a large text area). At the bottom, there are three buttons: 'Add Question', 'Confirm Module', and 'Cancel Construction'.



EditMod

Module Garage

Enter module name

Search For Question

Edit Question

Update Question or Delete Whole Question

Select Option Number

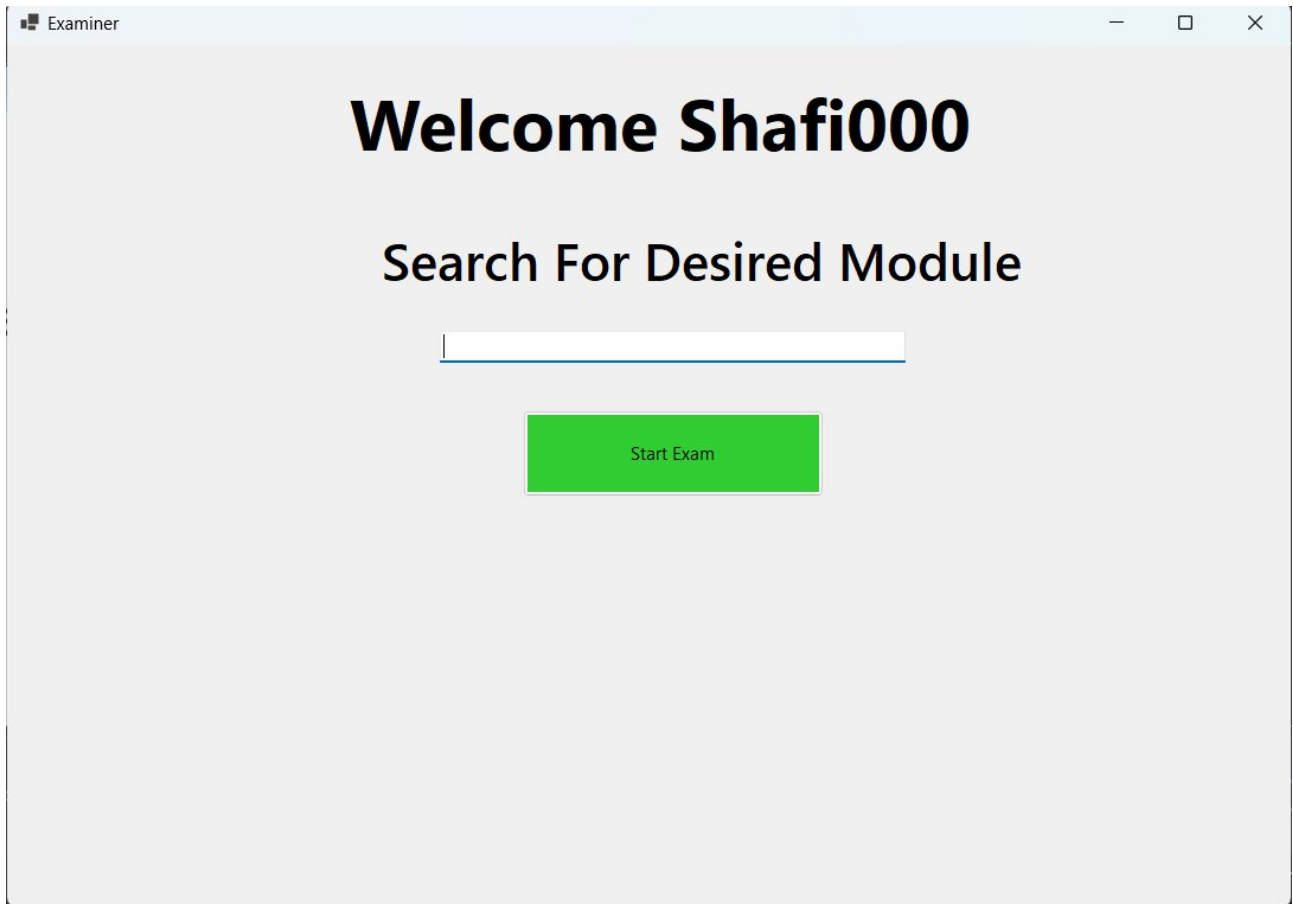
Edit Option to edit Update Option or Delete Option

Edit Short Explanation Update Explanation or Delete Explanation

Done Editing

This is the 'Module Garage' form. It has a title bar with 'EditMod'. The form is titled 'Module Garage' in bold. It contains several input fields and buttons: 'Enter module name', 'Search For Question', 'Edit Question', 'Update Question' or 'Delete Whole Question' (two buttons with 'or' between them), 'Select Option Number' (a dropdown menu), 'Edit Option to edit' (a text box), 'Update Option' or 'Delete Option' (two buttons with 'or' between them), 'Edit Short Explanation', 'Update Explanation' or 'Delete Explanation' (two buttons with 'or' between them), and 'Done Editing'.

Examiner Design

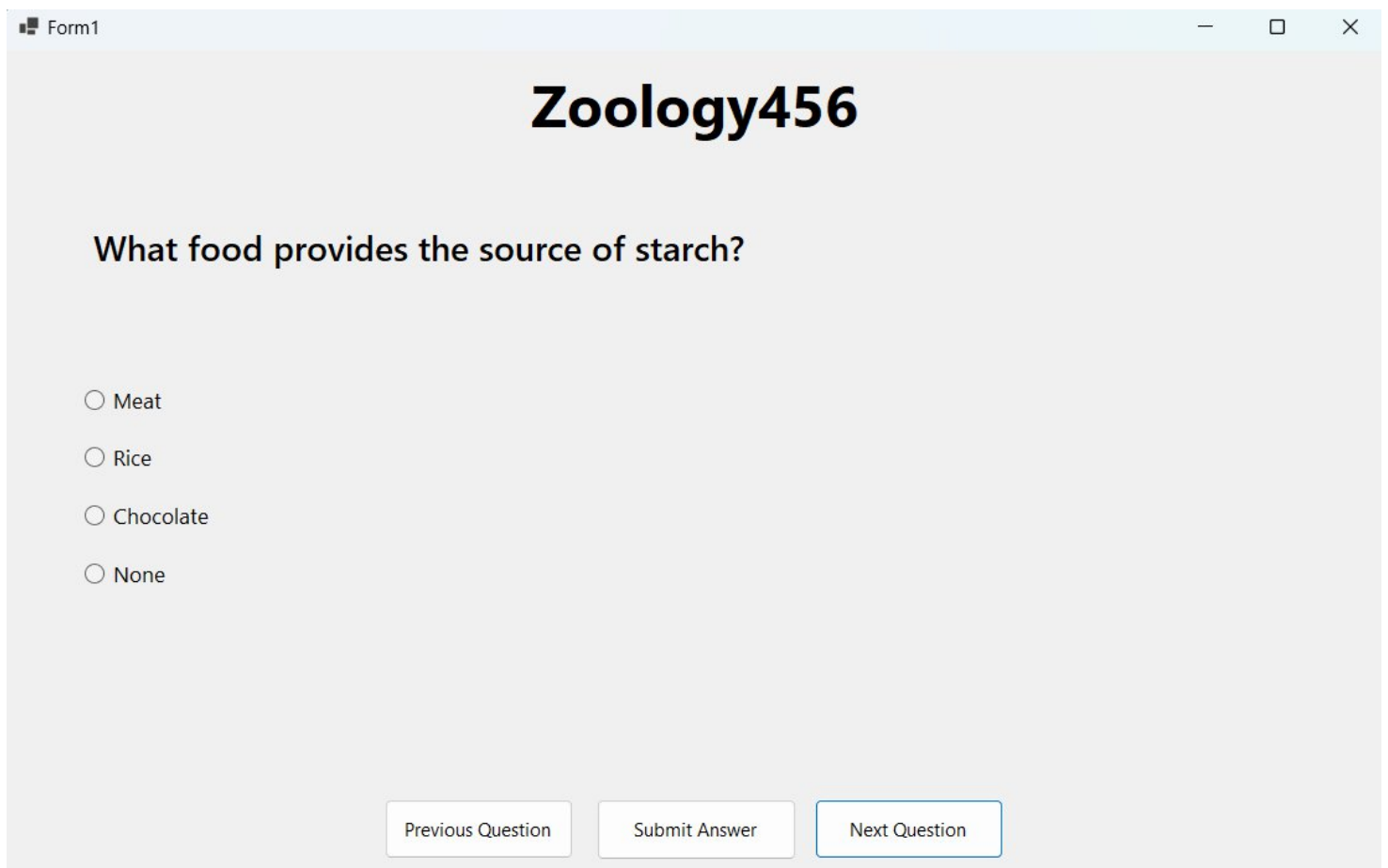


Examiner

Welcome Shafi000

Search For Desired Module

Start Exam



Form1

Zoology456

What food provides the source of starch?

☐ Meat

☐ Rice

☐ Chocolate

☐ None

Previous Question Submit Answer Next Question

Benefits

Centralized Question Bank: Educators and institutions can securely store, organize, and retrieve large volumes of multiple-choice questions in one dedicated system rather than scattered files.

Streamlined Test Assembly: The integrated construction module (`ConstructMod`) allows examiners to quickly pull from the vault to generate new exams without having to re-type or manually format questions.

Role-Based Access Control: With distinct portals for different users (`AuthorHP`, `ExaminerHP`) and secure authentication (`LoginPage`, `SignupPage`), the system ensures that only authorized personnel can view or alter sensitive exam content.

Efficient Revisions: The editing module (`EditMod`) allows for rapid corrections to questions or distractors (wrong options) without needing to overhaul entire test papers.

Automated Assessment: Delivering the test through the `Examination` module allows for immediate, automated grading, drastically reducing the manual workload on teachers and eliminating human grading errors.

Future Planning

Cloud Database Migration: Transitioning the local data storage to a cloud-hosted relational database (e.g., Azure SQL or AWS) to allow real-time, remote access for authors and students across different locations.

Advanced Analytics & Reporting: Implementing data visualization dashboards for examiners to track student performance trends, identify overly difficult questions, and analyze class averages.

Web or Mobile Integration: Upgrading the Windows desktop architecture to a web-based application or cross-platform mobile app to increase accessibility for students taking exams on diverse devices.

Anti-Cheating Mechanisms: Adding secure examination features, such as browser-lockdown modes, randomized question shuffling, or randomized option orders to maintain test integrity.

Multimedia Support: Upgrading the question construction tools to support image, audio, and video embeds within the MCQ prompts and options.

Conclusion

The MCQ Vault project successfully delivers a structured, role-based platform designed to modernize the management of multiple-choice examinations. By compartmentalizing question authoring, editing, and test execution into dedicated modules, the application significantly reduces the administrative burden on educational staff. With its robust foundational features including secure user authentication and dedicated examiner workflows the project provides a highly scalable solution that streamlines the traditional testing process and establishes a strong framework for future educational technology enhancements.